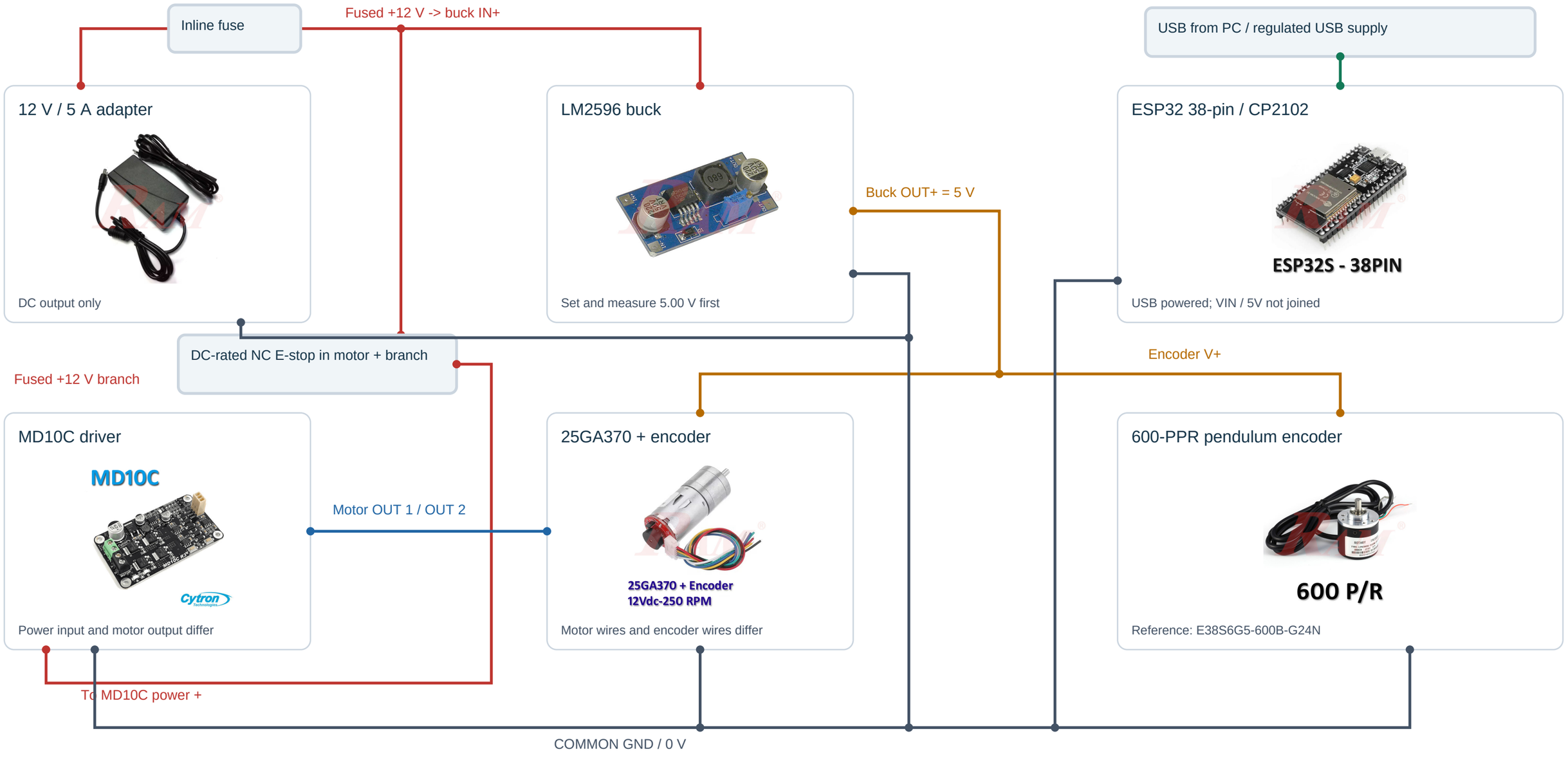
Power wiring with component photos

USB powers the ESP32. Buck output powers only encoders verified to accept 5 V. Signal wiring: page 2.



Power terminals and purpose

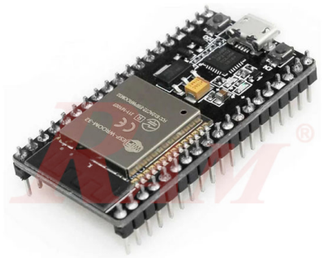
Crossings without dots are not joined; paired wires are labeled.

Connection	Purpose / rule
Adapter +12 V -> fuse -> split	One branch feeds buck IN+. The other feeds MD10C power + through the DC-rated E-stop. Encoder power can remain on when motor power is interrupted.
Common GND	Supply negative, driver power -, buck IN-/OUT-, ESP32 GND and both encoder grounds. Route motor return separately from signal returns.
Buck OUT+ / ESP32 USB	Buck OUT+ supplies verified 5 V encoders only. ESP32 uses USB. Do not tie buck 5 V to USB 5 V or apply 12 V to the ESP32.

Signal wiring and ESP32 pin assignments

Connections land on named terminals, not assumed locations in product photos. Use GPIO numbers in firmware.

ESP32 controller



GPIO logic is 3.3 V

GPIO27 -> PWM

GPIO26 -> DIR

GND -> GND

A -> GPIO32; B -> GPIO33

A -> GPIO18; B -> GPIO19

MD10C



PWM / DIR / signal GND

Motor A/B interface

3.3 V output: direct.

5 V push-pull: suitable buffer or designed dividers.

Open collector: 3.3 V pull-ups if supported.

A / B

Motor encoder



Verify supply and output stage

Pendulum A/B interface

Each A/B line has its own external pull-up to ESP32 3V3.

4.7 kohm is a starting value; verify rise time and LOW level.

No pull-up to 5 V / 12 V. See page 3.

A / B

Pendulum encoder



NPN open collector if model matches

Encoder connections
Both encoders share ESP32 GND. Supply connections are on page 1. Identify V+, GND, A and B from the actual device pinout.
Never infer motor or encoder wire functions from color alone.

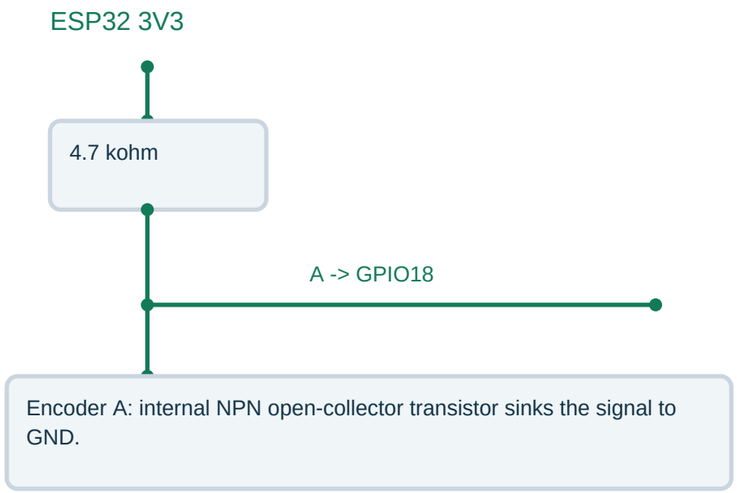
Pins used and why they are suitable

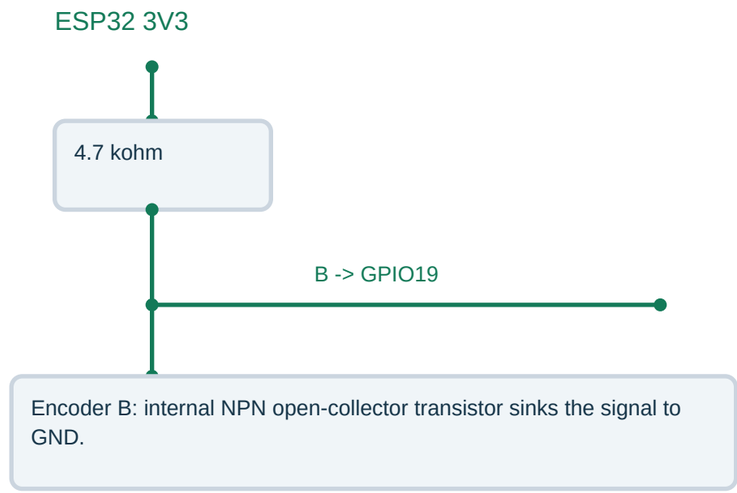
ESP32 pin	Destination	Reason / role
GPIO27	MD10C PWM	Output-capable GPIO supports LEDC PWM. Start around 10 kHz; MD10C supports up to 20 kHz. Hold LOW at startup.
GPIO26	MD10C DIR	Digital output selects direction. One H-bridge channel drives one motor forward and reverse.
GPIO32 / GPIO33	Motor encoder A / B	Input-capable pins support pulse counting / quadrature decoding. Apply verified 3.3 V signals.
GPIO18 / GPIO19	Pendulum encoder A / B	Input-capable pins support a second quadrature encoder. Reserve these pins; do not also assign SPI.
GND / 3V3	Common reference / pull-ups	GND is the signal reference; 3V3 feeds pull-ups. Selected GPIOs avoid ESP32 boot-strapping pins and flash pins GPIO6-11.

Encoder interface details and final checks

The motor encoder interface is conditional until its actual signal type and voltage are established.

600-PPR encoder: repeat this circuit for A and B





Encoder V+: buck 5 V for the RAM 5-24 V model. Pull-ups use 3.3 V, not encoder V+. Confirm there are no internal higher-voltage pull-ups. Verify HIGH and LOW levels with the actual unit. At 600 PPR, x4 decoding nominally gives 2,400 counts/revolution (0.15 degrees/count); confirm the vendor pulse convention.

Before connecting the motor encoder

If A/B is 5 V push-pull, use two suitable 5 V-tolerant buffer channels powered from 3.3 V or engineered resistor dividers. If open collector, use 3.3 V pull-ups only if supported. If true 3.3 V outputs, connect directly. Do not assume an Ampere motor has the RAM motor pinout, gearing or encoder specification. A multimeter averaging rotating pulses does not establish their HIGH voltage.

Commissioning and protection

1. Verify polarity, ground continuity and absence of supply shorts. Set buck voltage before attaching encoders.
2. Keep PWM LOW during reset and initialization; add a hardware PWM pull-down if the driver does not ensure a defined LOW. Require deliberate re-enable after E-stop.
3. Verify encoder counts and direction, then run the motor unloaded at low PWM. Check current, direction and E-stop interruption.
4. Size fuse, wire and DC switching contacts for actual current. Use secure terminals; keep motor current off solderless breadboards.
5. MD10C has regenerative braking and no adjustable motor-current limit. Check supply-voltage rise during reversal/braking; bulk capacitance alone may not absorb repeated returned energy. Add appropriate suppression / energy handling after measurement.

Photo credits and technical references - clickable links

ESP32 / RAM

MD10C / RAM

Motor / RAM (reference photo)

600-PPR encoder / RAM

Adapter / RAM

Buck / RAM

Cytron MD10C specifications

Espressif ESP32 datasheet

TI LM2596 specifications